

Scenario 03 - Developing a Hospital Management System (HMS)

Workflow Lists

1. To Do - All tasks before work begins.
2. Ongoing - Tasks actively being worked on.
3. Completed - Finished tasks.
4. Reopened - Tasks revisited due to bugs or issues.

Phase 1 - Planning and Requirement Gathering

Due Date: 22.12.2024

Main Task 1: Requirement Gathering

- Assigned to: Project Manager (PM)
- Priority: High
- Due Date: 19.12.2024
 - Subtasks:
 1. Meet with hospital stakeholders to understand the needs
 - Assigned to: PM
 - Priority: High
 - Due Date: 17.12.2024
 2. Create a list of required features (patient management, billing, appointment scheduling, etc.)
 - Assigned to: PM
 - Priority: High
 - Due Date: 18.12.2024
 3. Document functional requirements
 - Assigned to: PM
 - Priority: High
 - Due Date: 19.12.2024

Main Task 2: System Architecture Design

- Assigned to: Tech Lead
- Priority: High
- Due Date: *22.12.2024*
 - Subtasks:
 1. Design database schema for patient records and appointments
 - Assigned to: Senior Developer
 - Priority: High
 - Due Date: *20.12.2024*
 - Level 3 Subtasks:
 - 1.1. Define database tables for patient management
 - Assigned to: Junior Developer
 - Priority: High
 - Due Date: *19.12.2024*
 - 1.2. Set up relationships between tables (e.g., patients, doctors, appointments)
 - Assigned to: Junior Developer
 - Priority: High
 - Due Date: *20.12.2024*
 - 1.3. Perform initial database testing with sample data
 - Assigned to: QA Engineer
 - Priority: Medium
 - Due Date: *21.12.2024*
 2. Select technology stack (Frontend: React, Backend: Node.js, Database: PostgreSQL)
 - Assigned to: Tech Lead
 - Priority: High
 - Due Date: *21.12.2024*
 3. Define APIs for patient management and appointment scheduling
 - Assigned to: Senior Developer
 - Priority: Medium

- Due Date: *22.12.2024*
- Level 3 Subtasks:
 - 3.1. Create patient API endpoints
 - Assigned to: Junior Developer
 - Priority: High
 - Due Date: *21.12.2024*
 - 3.2. Create appointment scheduling API endpoints
 - Assigned to: Junior Developer
 - Priority: High
 - Due Date: *22.12.2024*
 - 3.3. Write API documentation for developers
 - Assigned to: Junior Developer
 - Priority: Medium
 - Due Date: *22.12.2024*

Phase 2 - Development

Due Date: *05.01.2025*

Main Task 1: Patient Management Module

- Assigned to: Tech Lead
- Priority: High
- Due Date: *02.01.2025*
 - Subtasks:
 - 1. Develop backend API for patient record management
 - Assigned to: Senior Developer
 - Priority: High
 - Due Date: *28.12.2024*
 - Level 3 Subtasks:
 - 1.1. Create API endpoint for adding patient records
 - Assigned to: Junior Developer
 - Priority: High

- Due Date: *26.12.2024*
- 1.2. Create API endpoint for updating patient records
 - Assigned to: Junior Developer
 - Priority: High
 - Due Date: *27.12.2024*
- 1.3. Test API endpoints for functionality
 - Assigned to: QA Engineer
 - Priority: Medium
 - Due Date: *28.12.2024*
- 2. Build frontend interface for patient records
 - Assigned to: Junior Developer
 - Priority: Medium
 - Due Date: *30.12.2024*
 - Level 3 Subtasks:
 - 2.1. Design patient management form (Name, Age, Medical History, etc.)
 - Assigned to: Junior Developer
 - Priority: Medium
 - Due Date: *29.12.2024*
 - 2.2. Integrate form validation and error handling
 - Assigned to: Junior Developer
 - Priority: Medium
 - Due Date: *30.12.2024*
- 3. Integrate backend with frontend
 - Assigned to: Senior Developer
 - Priority: High
 - Due Date: *01.01.2025*

Main Task 2: Appointment Scheduling Module

- Assigned to: Tech Lead
- Priority: High
- Due Date: *05.01.2025*

- Subtasks:
 - 1. Develop API for appointment creation and tracking
 - Assigned to: Senior Developer
 - Priority: High
 - Due Date: *02.01.2025*
 - Level 3 Subtasks:
 - 1.1. Create appointment creation endpoint
 - Assigned to: Junior Developer
 - Priority: High
 - Due Date: *30.12.2024*
 - 1.2. Add logic to prevent overlapping appointments
 - Assigned to: Junior Developer
 - Priority: Medium
 - Due Date: *31.12.2024*
 - 1.3. Test appointment creation API for edge cases
 - Assigned to: QA Engineer
 - Priority: Medium
 - Due Date: *02.01.2025*

Phase 3 - Testing & Bug Fixing

Due Date: *15.01.2025*

Main Task 1: Unit Testing

- Assigned to: QA Lead
- Priority: High
- Due Date: *10.01.2025*
 - Subtasks:
 - 1. Test patient management API endpoints
 - Assigned to: QA Engineer
 - Priority: High
 - Due Date: *07.01.2025*

- Level 3 Subtasks:
 - 1.1. Validate adding patient records API
 - Assigned to: QA Tester
 - Priority: High
 - Due Date: *06.01.2025*
 - 1.2. Validate updating patient records API
 - Assigned to: QA Tester
 - Priority: High
 - Due Date: *07.01.2025*
 - 1.3. Test error scenarios for invalid inputs
 - Assigned to: QA Tester
 - Priority: Medium
 - Due Date: *07.01.2025*
- 2. Test appointment scheduling API endpoints
 - Assigned to: QA Engineer
 - Priority: High
 - Due Date: *09.01.2025*
 - Level 3 Subtasks:
 - 2.1. Validate appointment creation API
 - Assigned to: QA Tester
 - Priority: High
 - Due Date: *08.01.2025*
 - 2.2. Test overlapping appointment prevention logic
 - Assigned to: QA Tester
 - Priority: High
 - Due Date: *09.01.2025*
 - 2.3. Test edge cases for appointment modifications
 - Assigned to: QA Tester
 - Priority: Medium
 - Due Date: *09.01.2025*

Main Task 2: Integration Testing

- Assigned to: QA Lead
- Priority: High
- Due Date: *12.01.2025*
 - Subtasks:
 1. Verify API integration with frontend components
 - Assigned to: QA Engineer
 - Priority: High
 - Due Date: *11.01.2025*
 - Level 3 Subtasks:
 - 1.1. Test patient management module integration
 - Assigned to: QA Tester
 - Priority: High
 - Due Date: *10.01.2025*
 - 1.2. Test appointment scheduling module integration
 - Assigned to: QA Tester
 - Priority: High
 - Due Date: *11.01.2025*
 - 1.3. Verify error handling in API responses
 - Assigned to: QA Tester
 - Priority: Medium
 - Due Date: *11.01.2025*
 2. Validate user interactions in the UI
 - Assigned to: QA Engineer
 - Priority: Medium
 - Due Date: *12.01.2025*

Main Task 3: Bug Fixing

- Assigned to: Tech Lead
- Priority: High
- Due Date: *15.01.2025*
 - Subtasks:
 1. Fix API-related bugs

- Assigned to: Senior Developer
- Priority: High
- Due Date: *14.01.2025*
- Level 3 Subtasks:
 - 1.1. Resolve database connectivity issues
 - Assigned to: Junior Developer
 - Priority: High
 - Due Date: *13.01.2025*
 - 1.2. Fix incorrect API responses for appointment creation
 - Assigned to: Junior Developer
 - Priority: High
 - Due Date: *14.01.2025*
 - 1.3. Perform retesting after bug fixes
 - Assigned to: QA Engineer
 - Priority: Medium
 - Due Date: *15.01.2025*

2. Fix UI-related bugs

- Assigned to: Senior Developer
- Priority: High
- Due Date: *15.01.2025*
- Level 3 Subtasks:
 - 2.1. Fix validation issues in the patient form
 - Assigned to: Junior Developer
 - Priority: Medium
 - Due Date: *14.01.2025*
 - 2.2. Correct appointment module UI alignment
 - Assigned to: Junior Developer
 - Priority: Medium
 - Due Date: *15.01.2025*

Phase 4 - Deployment & Training

Due Date: *25.01.2025*

Main Task 1: Deployment

- Assigned to: Tech Lead
- Priority: High
- Due Date: *20.01.2025*
 - Subtasks:
 1. Set up production server environment
 - Assigned to: Senior Developer
 - Priority: High
 - Due Date: *17.01.2025*
 2. Deploy backend APIs to the server
 - Assigned to: Senior Developer
 - Priority: High
 - Due Date: *18.01.2025*
 3. Deploy frontend application to the server
 - Assigned to: Tech Lead
 - Priority: High
 - Due Date: *19.01.2025*
 4. Perform smoke testing on the production server
 - Assigned to: QA Lead
 - Priority: Medium
 - Due Date: *20.01.2025*

Main Task 2: Training Hospital Staff

- Assigned to: Project Manager
- Priority: Medium
- Due Date: *25.01.2025*
 - Subtasks:
 1. Prepare training materials (guides, videos, and presentations)
 - Assigned to: PM
 - Priority: Medium
 - Due Date: *21.01.2025*

2. Conduct training sessions for hospital staff
 - Assigned to: PM and QA Lead
 - Priority: Medium
 - Due Date: *23.01.2025*
3. Collect feedback from staff and provide clarifications
 - Assigned to: PM
 - Priority: Low
 - Due Date: *25.01.2025*